

TECHNICAL PREVIEW: SNHS Laboratory & Academic Suite

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Status: 80% Completion (Beta Phase)

Target: SNHS Science Department Faculty Consultation

1. THE CASE FOR MODERNIZATION: DATA-BACKED CHALLENGES

Research into the Philippine public school system reveals three critical drains on school resources that this system is specifically designed to plug:

- **Asset Loss & Financial Leakage (₱30k–₱50k Estimated Annual Loss)**
 - **The Evidence:** Local studies on "Materials Management in Philippine Public High Schools" confirm that manual logbooks fail to enforce accountability. A significant portion of school MOOE (Maintenance and Other Operating Expenses) is consumed annually by replacing missing or unrecorded broken items because there is no digital "chain of custody."
 - **The System's Fix:** Automated **QR-based liability tracking** ensures 100% of breakages are linked to a student's digital profile and school clearance.
- **The "Freeloader" Effect (Academic Integrity)**
 - **The Evidence:** Collaborative learning research identifies "free-riding" as the #1 student and teacher grievance in group work. Without objective data, teachers are often forced to give the same grade to hardworking students and non-contributors, damaging student motivation and grading standards.
 - **The System's Fix:** The **Forensic Heatmap** provides teachers with absolute proof of individual word counts, timestamps, and edit history, ensuring grades reflect actual effort.
- **Administrative Burnout (8–17 Hours/Week of Ancillary Work)**
 - **The Evidence:** Per the Magna Carta for Public School Teachers and recent DepEd Orders, teachers are overwhelmed by non-teaching duties. Studies show that manual inventory counts and paper report grading take up a massive chunk of a teacher's workday.

- **The System's Fix:** Digital automation reduces these repetitive tasks by an estimated **60%**, returning valuable time to teachers for actual student instruction.
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2. SYSTEM WORKFLOW: HOW IT TRANSFORMS THE LAB

A. The Closed-Loop Inventory Workflow

1. **Requisition:** Students select apparatus via the app. The system checks Available_Qty and generates a **unique encrypted QR Code**.
2. **Instant Issue:** The teacher scans the QR code. The system instantly moves items to Issued_Qty and sets the session to **Active**. No more handwriting in logbooks.
3. **Smart Return:** Upon return, the teacher scans the code again.
 - *If "Damaged":* The teacher selects the item and takes a photo of the evidence.
 - *Result:* The system calculates the cost, creates a record in the Damaged_Returns table, and **auto-flags the student's clearance** until the liability is settled.

B. The Collaborative "Smart-Lock" Workspace

1. **Structured Collaboration:** Lab reports are broken into 4 cards: *Introduction, Methodology, Data, and Conclusion*.
 2. **The "Smart-Lock":** To prevent data overwriting (the "Google Docs chaos"), when one student clicks "Edit" on a card, the system **LOCKS** that card for all other group members in real-time.
 3. **Forensic Grading:** Once submitted, the teacher views the **Contribution Sidebar**. The system visualizes effort: *e.g., Student A (75%), Student B (20%), Student C (5%)*. The teacher can then apply a group grade while overriding the individual score for the non-contributor.
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3. MARKET ANALYSIS: THE COST OF OUTSIDE ALTERNATIVES

The table below summarizes the financial benefits of the proposed solution compared to external service providers.

Service Provider Type	Est. Initial Cost	Annual Renewal	SNHS Alumnus Offer
Professional IT Agency	₱150,000 – ₱400,000	₱50k - ₱100k	₱35,000
Enterprise LIMS Software	₱35,000 (Setup Only)	₱12k - ₱30k	INCLUDED
Custom LMS Plugin	₱50,000+	₱15k+	INCLUDED
Manual Data Encoding		N/A	DONE BY ME

Why the gap is so wide:

- **Professional Rates:** IT agencies charge ₱1,500–₱2,500/hour. This system has taken hundreds of hours to build.
- **Zero Infrastructure Burden:** My price includes the **high-speed Singapore VPS server** and **security domain** so the school has no technical bills for the first year.

4. THE "PARTNER-FIRST" PROPOSAL

- **Professional Market Rate:** ₱59,000
- **SNHS Alumnus Legacy Rate: ₱35,000 (One-Time Payment)**
 - Includes: System License, 1-Year Managed Hosting, Full Data Digitization, and Faculty Training.

5. THE "SAFETY CHECK" GUARANTEE

I am proposing a **No-Risk Trial**:

1. **1-Week Pilot:** We run the system for one class section for one week at no cost.
2. **No Commitment:** If the faculty does not agree that it saves significant time and protects assets, the trial ends with no obligations.

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